

Stereographic projections - stereonet

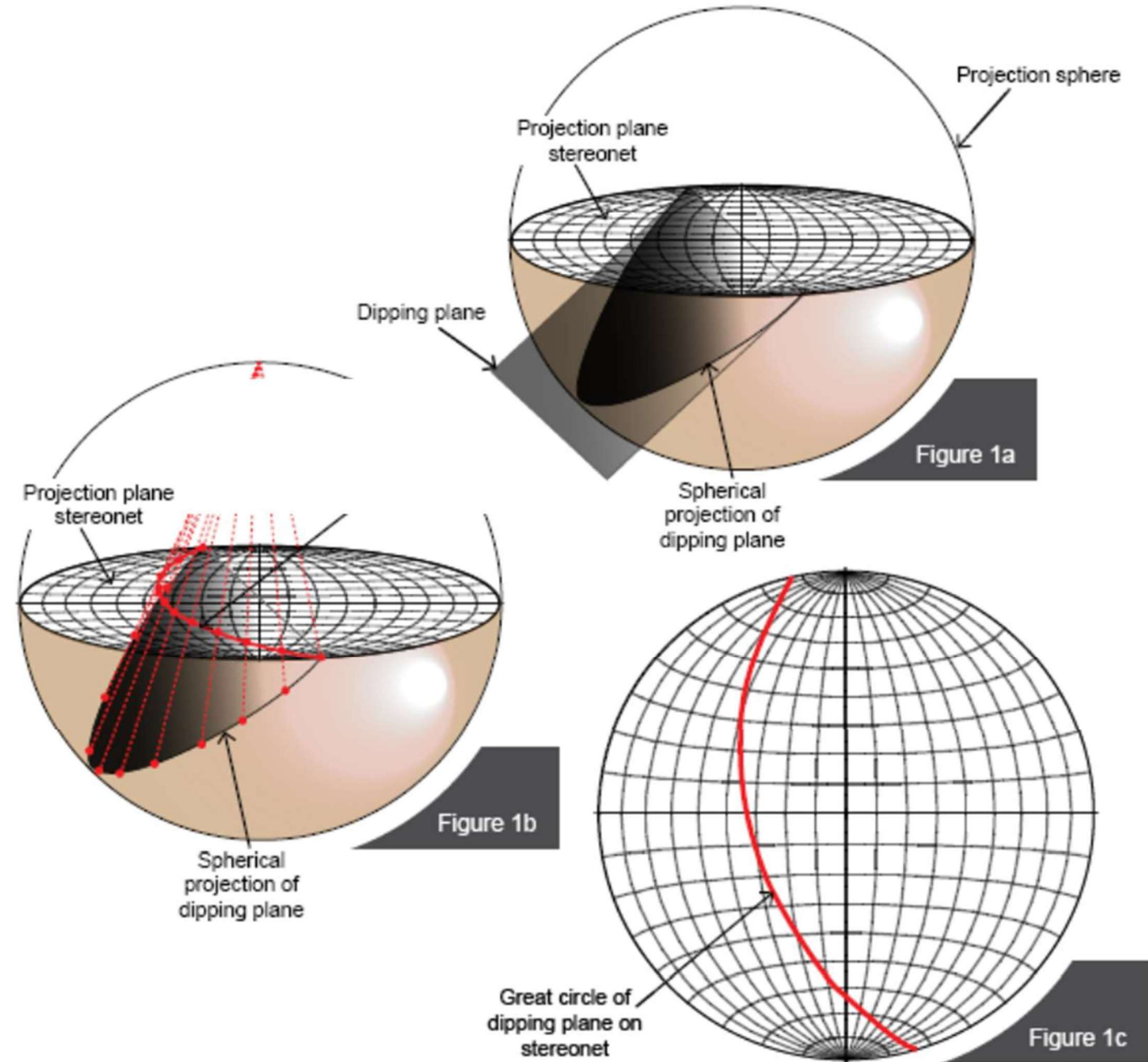
Engineering Geology Lab & Geological Engineering Lab

CVP121 & CVP 2401

- Dr. Deepanshu Shirole

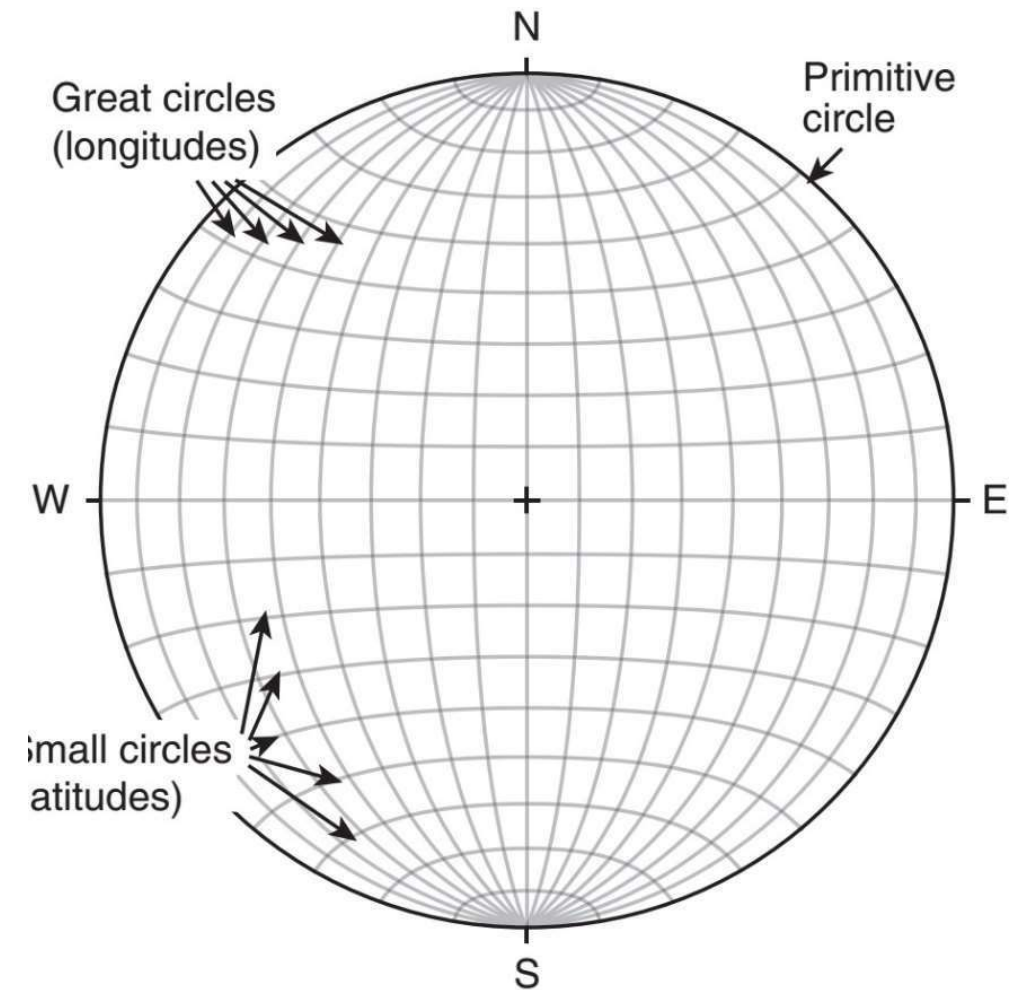
Aims of this LAB

- Plotting structural data on stereonet



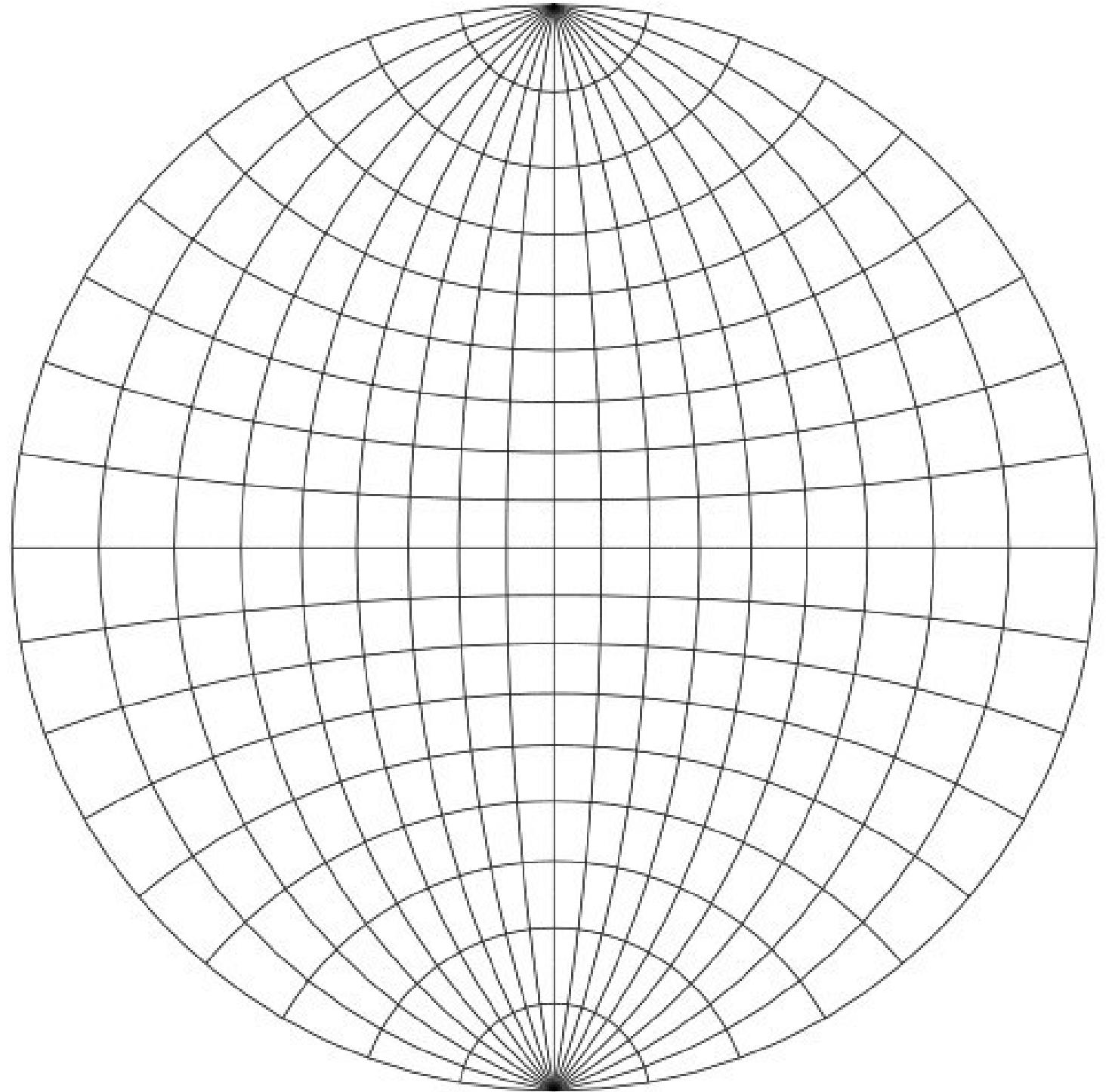
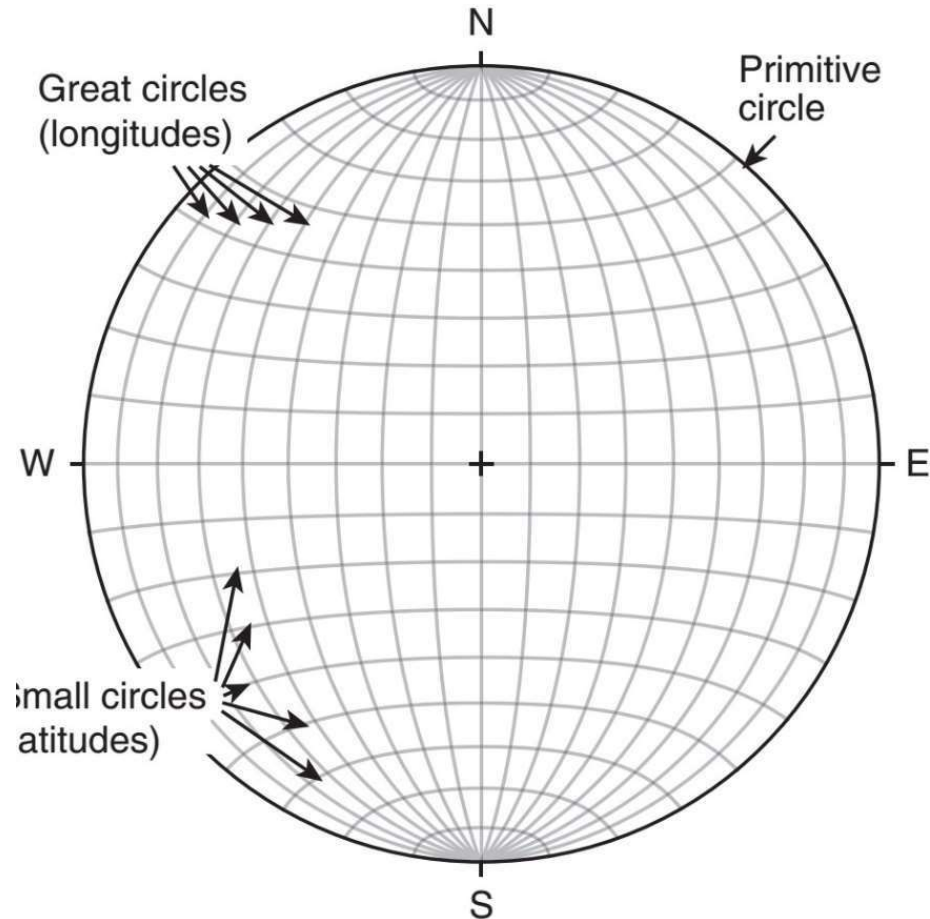
Concept of stereonet

- Stereographic projections are used to display orientation data.
- Stereonet can also be used in complex calculations such as rotations, calculating lines of intersections, etc.
- Stereonet looks like a globe or map of the entire earth. There are great circles (projections), **just like lines of longitude running from pole to pole**, and small circles (projections), **like lines of latitude that run across the globe from east to west**.

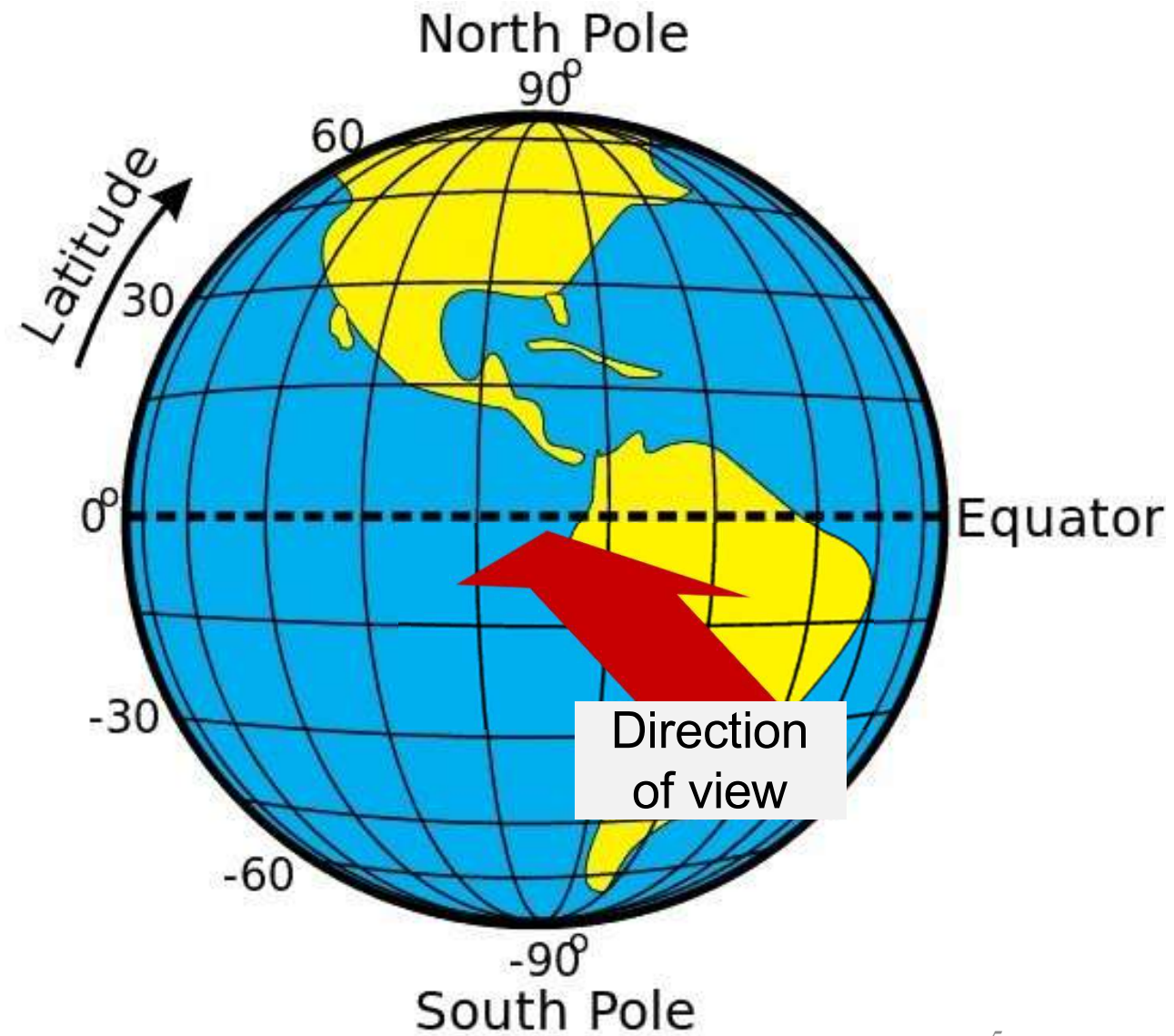
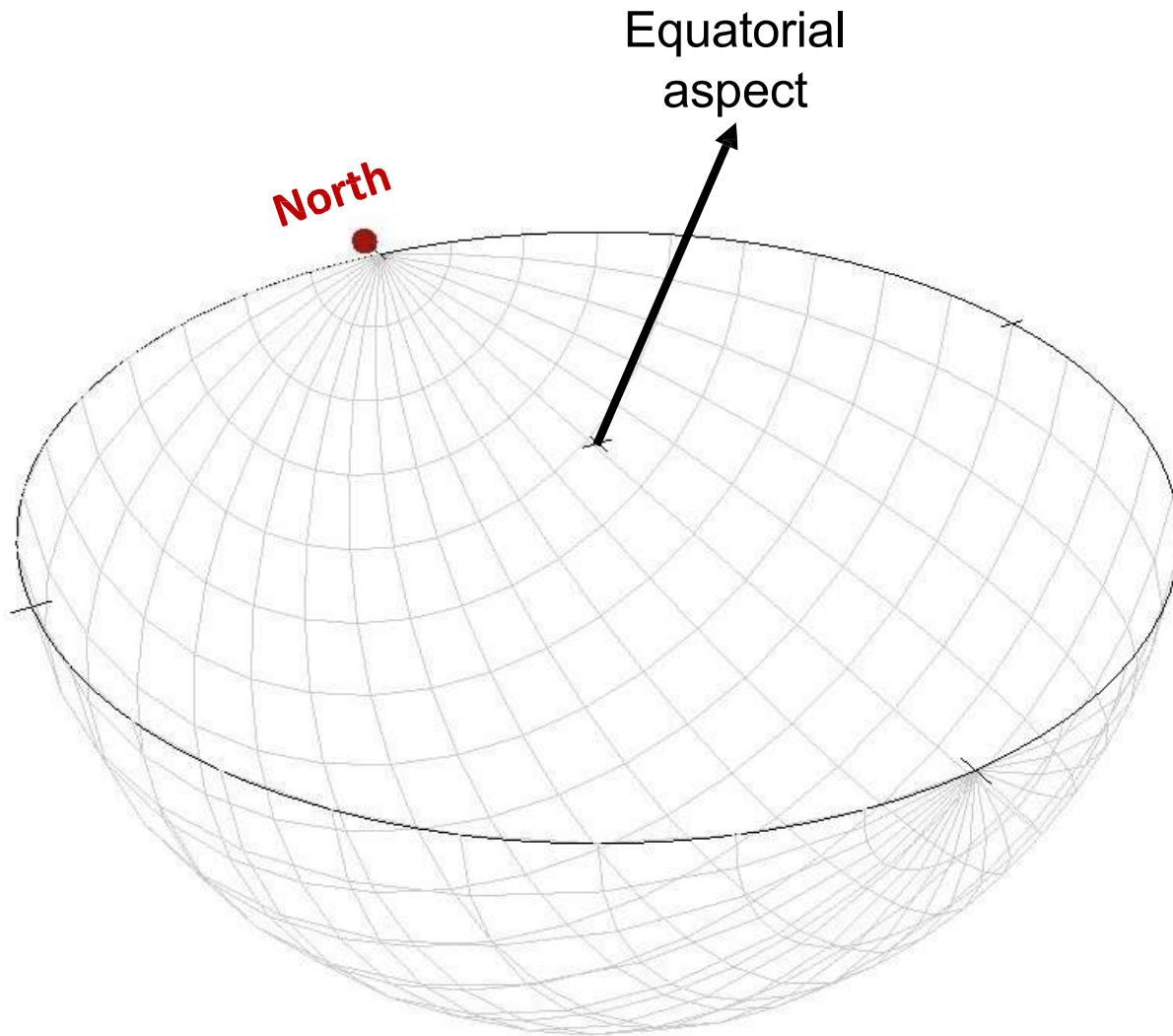


Stereonet (cont.)

- Used to plot structural data.

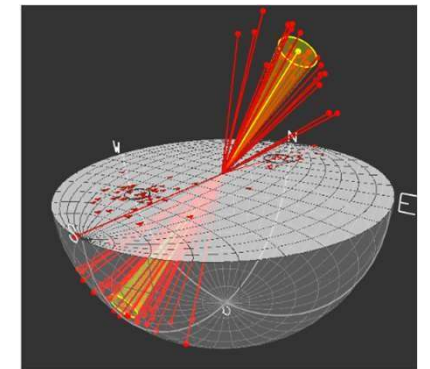
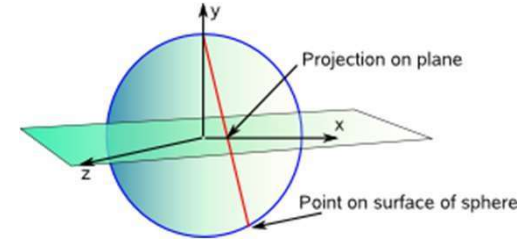
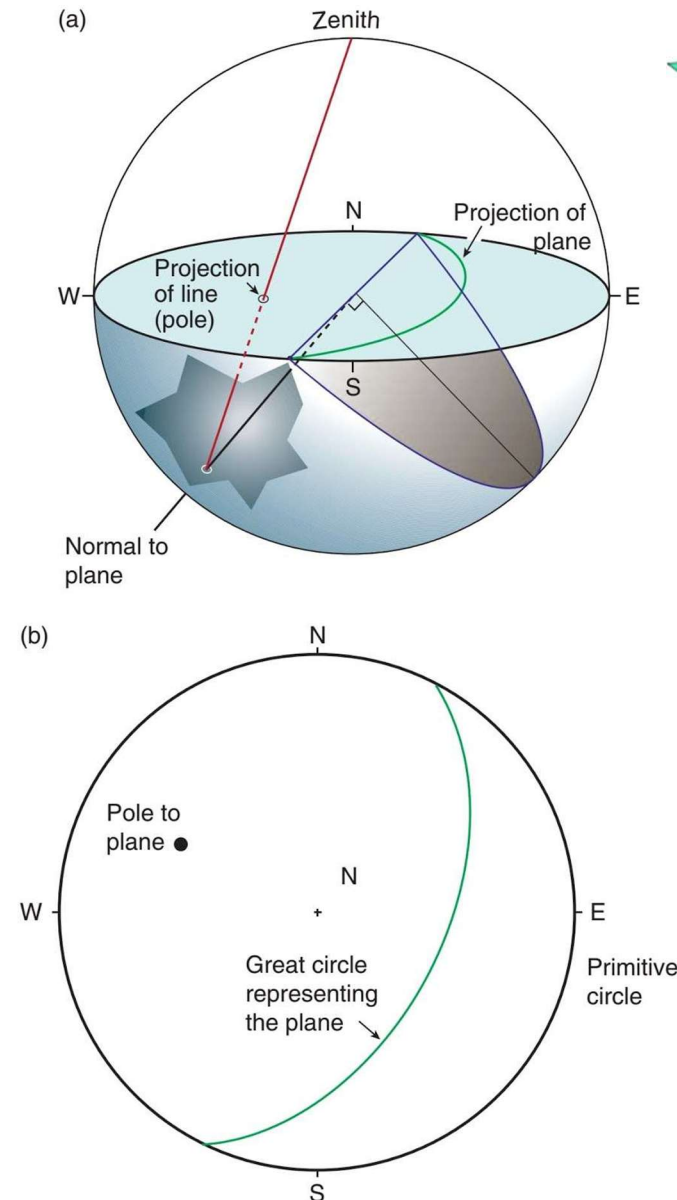


Equatorial aspect



Principles of stereographic projection

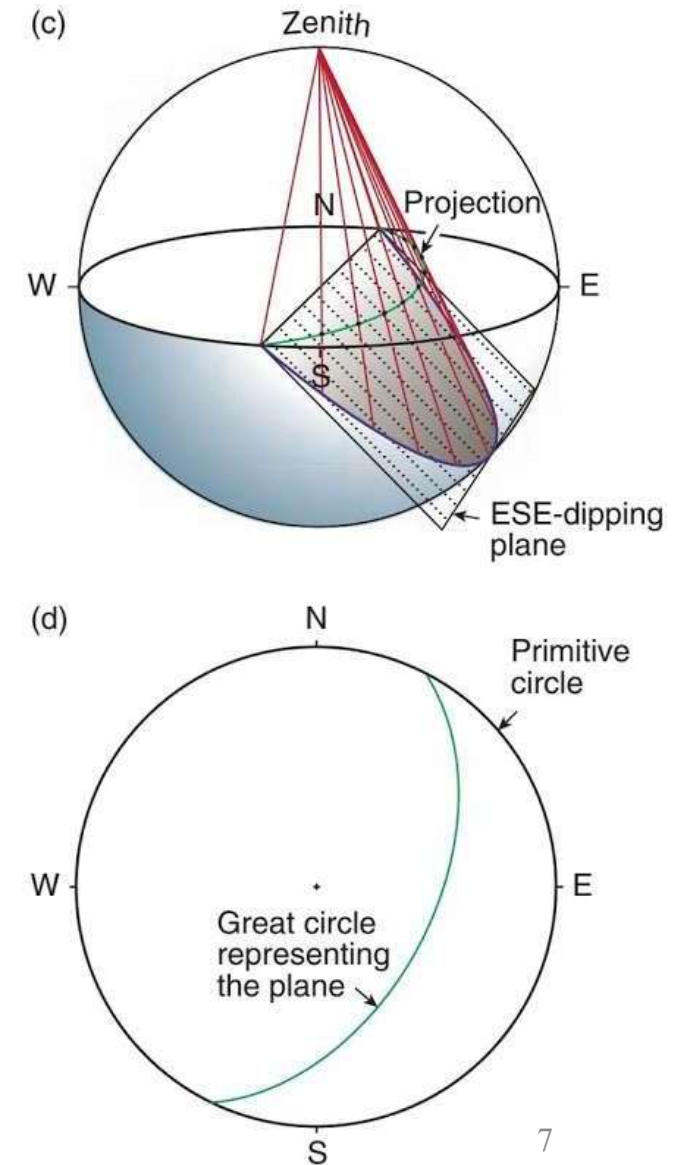
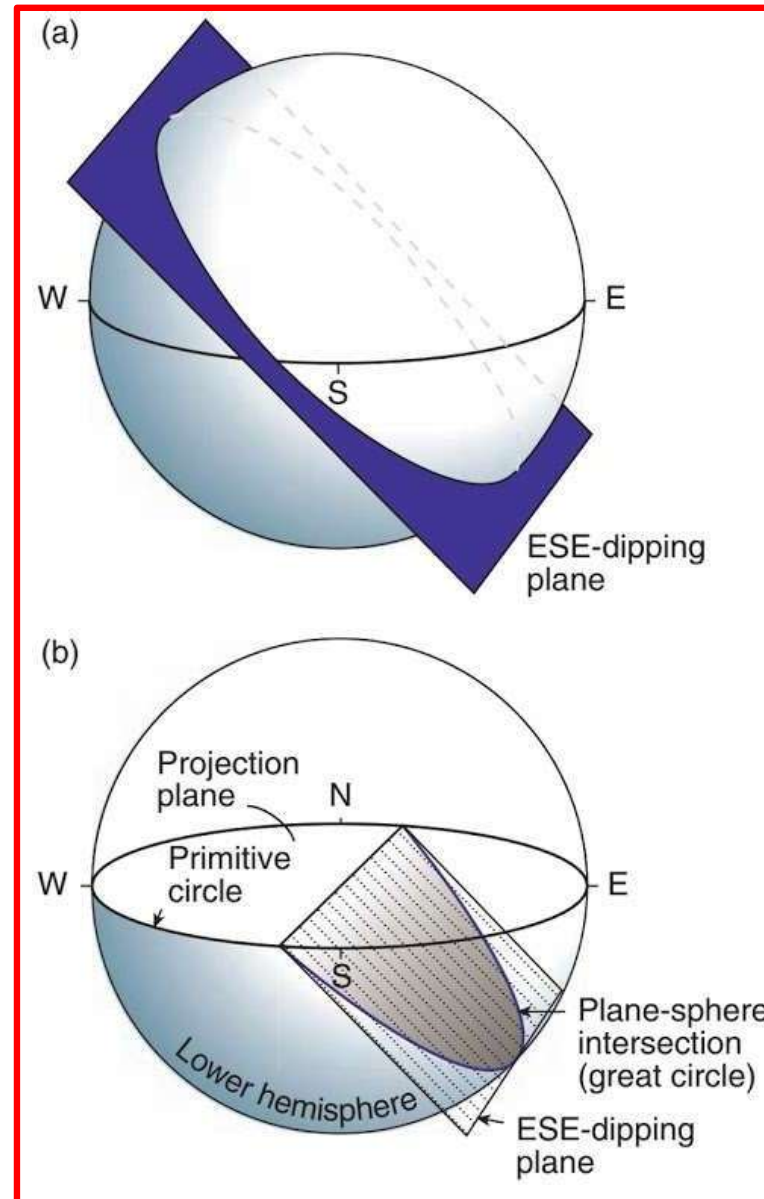
- Stereographic projections are primarily used to present planar and linear features in a 2D diagram.
- The projection plane is an imaginary horizontal plane passing through the center of sphere.



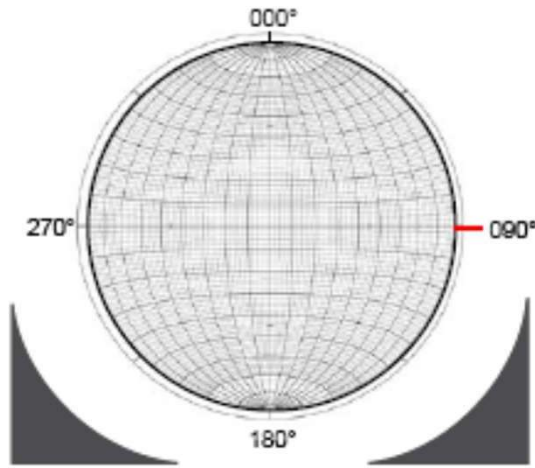
<http://www.geo.cornell.edu/geology/faculty/RWA/programs/stereonet-mobile.html>

Projection of a plane

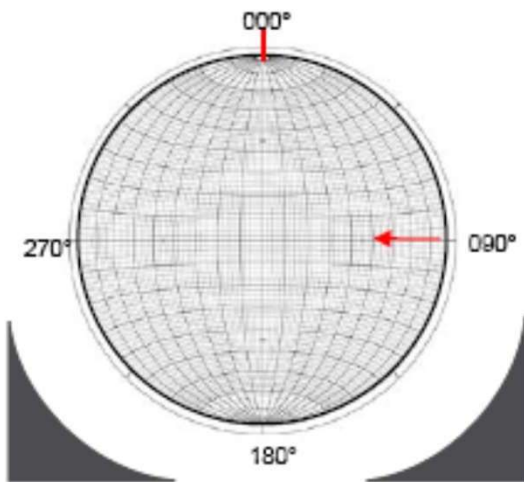
- Imagine a dipping or inclined plane, bedding or foliation pass through the center of sphere
- The plane intersects the lower hemisphere at a curved line and produce a plane representing the intersection plane between the dipping plane and the sphere
- The task is to project the intersection plane on projection plane



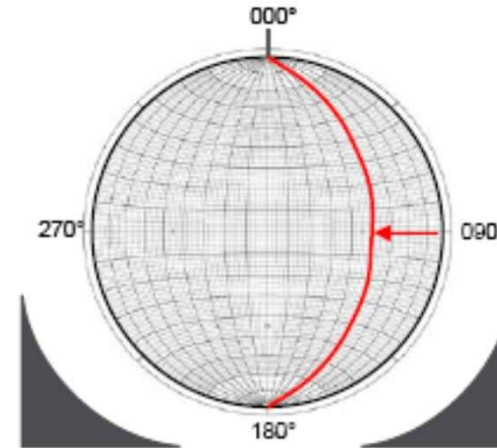
Plotting on stereonet: strike/dip 090/40S



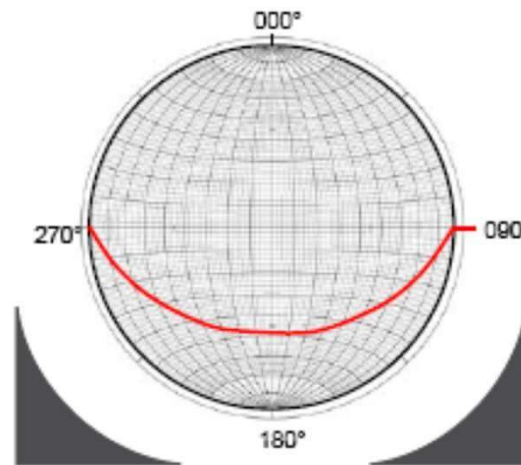
1. Mark on the strike - 090°
Note which way the plane is dipping, then rotate the tracing paper round until this mark is aligned with north on the stereonet.



2. Find the great circle of the plane by counting in the angle of dip along the equator from the primitive. Count in from the direction of dip as marked on the tracing paper (in this case S). Mark with a dot.

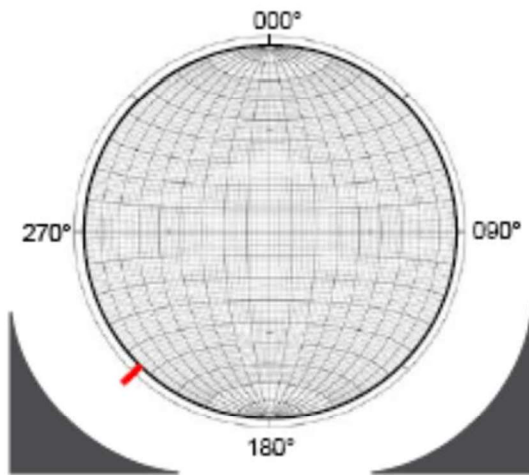


3. Using a sharp pencil or colour pencil, draw in the great circle from the north pole through point to the south pole.



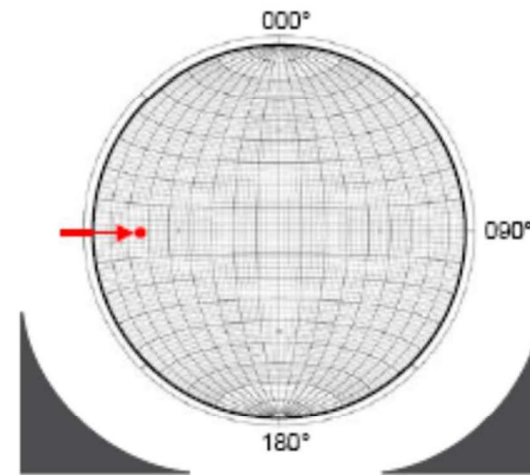
4. Rotate the tracing paper back to north. Check the plane is dipping in the correct direction and admire your work.

Plot a linear feature - (25/225)

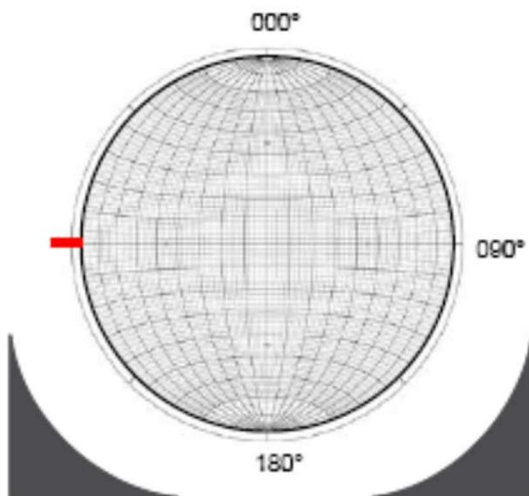


- Plunge and azimuth: 25/225

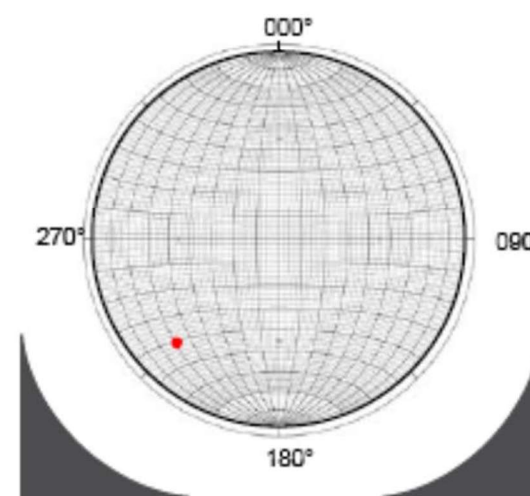
- Mark on the azimuth reading 225°



- From the primitive count in the plunge 25°. Mark on the lineation



- Rotate the tracing paper round until this mark is aligned with equator on the stereonet.



- Rotate the tracing paper back and admire your work.

- Thank you !
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